EMI CERVANTES

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EDUCATION

THE PENNSYLVANIA STATE UNIVERSITY | University Park, PA **Aug 2024 - May 2029 (Expected)**

Doctor of Philosophy in Statistics

Dissertation Committee: Samuel Baugh (chair), Matthew Beckman (co-chair), Neil Hatfield, Puiwa Lei

Qualifying Exam: PASSED - May 2025; Comprehensive Exam: May 2027 (expected)

Research Interests: Bayesian hierarchical modeling, latent variable modeling, longitudinal data analysis, attitudes and motivation.

UNIVERSITY OF CALIFORNIA, IRVINE | Irvine, CA **Sep 2020 - Jun 2024**

Bachelor of Science in Mathematics

HONORS & AWARDS

Penn State University

University Graduate Fellowship & Dean's Distinguished Graduate Fellowship, 2024-2029 (total award: $32,000 stipend per year).

Science Achievement Graduate Fellowship, 2026-2029 (total award: $37,000 stipend per year).

Innovation Fellowship, Penn State Office for Innovation, 2026 (total award: $5,000).

UC Irvine

UC Irvine Department of Mathematics Undergraduate Research and Department Service Award, 2024.

Pi Mu Epsilon Joint Mathematical Meetings Undergraduate Student Travel Grant, 2024.

Undergraduate Research Opportunities Program (UROP) Research Fellowship, 2023 (total award: $444).

UC Irvine School of Physical Sciences Advancing Inclusive Excellence (AIE) Award, 2023 (total award: $500).

Working Memory and Plasticity Symposium Best Presentation, 2023.

Career Pathways for Research in Learning, Education, Analytics, and Data Science (CP-LEADS) Fellowship, 2023 - 2024 (total award: $6,000, IES-funded).

SoCal Data Science Fellowship, 2022 (total award: $5,000, NSF-funded, REU).

Dean's Honor List, 2020 - 2024.

Hispanic Scholarship (HSF) Scholar, Walmart Foundation/HSF Scholarship Program, 2020 (total award: $4,000).

RESEARCH EXPERIENCE

A Longitudinal Bayesian Item Response Theory Model for Predicting Student Learning Progression

Penn State Department of Statistics | PI: Samuel Baugh, PhD | 2025 - Current

- Formulating a new *Bayesian Item Response Theory model* and implementing it with R and Stan to improve instructor feedback on student learning.
- Developing comprehensive data-processing and visualization pipelines to translate model outputs into data-informed feedback for instructors.

Impact of Comparative Judgement on Engagement and Affect in Large Statistics Classroom

Penn State Department of Statistics | PI: Matthew Beckman, PhD | 2026 - Current

- Designing a qualitative study to examine the effects of different active learning strategies on students' engagement and affect in a large-enrollment introductory statistics course.
- Collaborating with participants from different institutions, including three domestic universities and one international university, on data collection.

Causes of Math Anxiety: Mathematical Background and Attitudes

UC Irvine School of Education | PI: Lindsey Richland, PhD, Fernando Rodriguez, PhD | 2023 - 2024

- Analyzed large-scale educational data using *structural equation modeling* and *causal inference* in R to evaluate the impact of learning interventions across pre-post experimental designs.
- Designed and validated survey instruments to assess student cognitive and affective factors, directing multi-site data collection efforts and conducting routine exploratory data analysis.

Greedy Approach to Kaczmarz Method with Oblique Projection

UC Irvine Department of Mathematics | PI: Anna Ma, PhD | 2022 - 2024

- Developed and proved theorems for the new iterative algorithm, the *greedy Kaczmarz method with oblique projections (GKO)*, to solve large-scale linear systems.
- Utilized *Python* to run extensive numerical experiments of the algorithm to ensure its functionality and wrote the manuscript for publication.

Impact of Mediterranean and DASH Diets on MPN Symptoms

UC Irvine School of Medicine | PI: Angela Fleischman PhD, MD | 2022

- Employed a *linear mixed-effects model* to clinical trial data with R software to determine the effectiveness of Mediterranean diets in improving symptoms of patients with myeloproliferative neoplasms (MPN).
- Collaborated with a diverse group of researchers, including biologists and computer scientists, to address our research question and presented findings at conferences.

PROFESSIONAL EXPERIENCE

Data Analyst | Penn State Eberly College of Science **Oct 2026 – Present**

- Extract, integrate, and analyze multi-source institutional data to produce reports for college leadership, informing program assessment and policy decisions.
- Partner with departmental leaders and IT teams to design new survey tools, build secure data repositories, and provide accessible data visualization platforms to non-technical stakeholders on interpreting metrics to improve student outcomes at Penn State.

Research Assistant | Science of Learning Lab at UC Irvine **Sep 2023 – Jun 2024**

- Conducted exploratory data analysis on active study data in R, presenting key trends and findings to guide weekly lab discussions.
- Led multiple in-person studies across elementary and middle schools in Orange County, CA, to study children's higher-order thinking skills in math education.

PROJECTS

Topic Modeling using Mean-Field Variational Inference and Normalizing Flows 2026

- Validated theoretical upper bounds of convergence for *Latent Dirichlet Allocation* with *mean-field variational inference*. Implemented the algorithm using *Python* on the 20 Newsgroups dataset consisting of 18,000 documents.
- Incorporated *discrete and mixed normalizing flows* into the variational inference framework to test theoretical bounds.

Anime Genre Predictor and Recommendation Engine 2021

- Developed a predictive model using *Scikit-learn LogisticRegression* with *Python* to forecast anime genres based on reviews and airing time.
- Implemented an anime recommendation system with a *cosine similarity algorithm*, providing personalized anime recommendations based on user preferences and viewing patterns

PRESENTATIONS

Talks

Cervantes E. (2026 July). Facilitating Student-Instructor Conversation with Bayesian Item Response Theory. Presented at 12th International Conference on Teaching Statistics. Brisbane, QLD, Australia.

Cervantes E. & Tran J. (2024 January). A Greedy Approach to Kaczmarz Method with Oblique Projections. Presented at Joint Mathematics Meetings. San Francisco, CA.

Poster Presentations

Cervantes E. (2026 August). A Longitudinal Bayesian Item Response Theory Model for Predicting Student Learning Progressions. Presented at Joint Statistical Meetings. Boston, MA.

Cervantes E. (2024 May). The Complex Relationship Between Math Anxiety and Mind Wandering. Presented at Annual Undergraduate Research Symposium. Irvine, CA.

Cervantes E. & Tran J. (2024 April). A Greedy Approach to Kaczmarz Method with Oblique Projections. Presented at Southern California Applied Mathematics Symposium. San Diego, CA.

Cervantes E. & Ma S. (2023, June). Effectiveness of Worked Examples on Reducing Math Anxiety. Presented at Working Memory and Plasticity Lab Symposium. Irvine, CA.

Cervantes E., Davityan M., Halabi A., Truong J., Wu M. (2022, August). Impact of Mediterranean and DASH Diets on MPN Symptoms. Presented at Project RAISE Summer Research Symposium. Fullerton, CA.

PUBLICATIONS

Cervantes E., Baugh S., Hatfield N. (2026+). A Longitudinal Bayesian Item Response Theory Model for Predicting Student Learning Progression. Manuscript in Preparation.

TEACHING

Penn State University (Graduate Teaching Assistant)

STAT 100: Statistical Concepts and Reasoning **Fall 2026**

- 4 lab sections of ~45 students each

STAT 200: Elementary Statistics **Spring 2026**

- 2 lab sections of ~85 students each

STAT 380: Data Science Through Statistical Reasoning and Computation **Fall 2025**

SERVICE & LEADERSHIP

Penn State University

Statistics Graduate Student Association 2025 - Current

- Historian (2025-2026): Managed social media platforms, including the website and Instagram.
- Vice President (2026-Current): Organize events, including workshops and social events for Statistics graduate students at Penn State.

Graduate Outreach Committee, Statistics Department (chair) 2026 - Current

- Lead a group of graduate students in planning local science outreach events, including workshops and Florence Nightingale Day.

Climate & Diversity Subcommittee, Eberly College of Science 2025 - 2027

- Plan and evaluate college-wide initiatives to enhance the inclusion and professional well-being of graduate students and faculty in the Penn State Eberly College of Science as the Department of Statistics Graduate Student Representative.

UC Irvine

Data @ UCI 2022 - 2024

- Vice President of Logistics (2022-2023): Coordinated all aspects of event logistics, including venue selection and vendor management.
- Executive Vice President (2023-2024): Facilitated meetings with industry and university executives to discuss potential partnerships and collaborations for weekly events and the inaugural Datathon, a Data Science hackathon.

Undergraduate Mathematics Committee (Co-founder and President) 2023 - 2024

- Collaborated effectively with board members to organize and host weekly professional and social events, including professor research seminars, panels, and UC Irvine's first inaugural integration bee.

Faces in Mathematics 2021 - 2023

- External Vice President (2022-2023): Managed study room for members and hosted workshops to support lower-division math majors in constructing their four-year college plan.
- Director of Professional Development (2021-2022): Established and maintained connections with industry and the Department of Mathematics at UC Irvine for conferences and weekly meetings.

SKILLS

Computer

R (advanced): tidyverse, lme4, laavan, Quarto, mirt, stan
Python (advanced): NumPy, pandas, Matplotlib, scikit-learn, PyTorch, TensorFlow
SQL (intermediate): MySQL, schema design, data querying, filtering & aggregations, data manipulation
MATLAB (intermediate): Scientific computing, scripting, data visualization
LaTeX (advanced): Document preparation, Beamer, bibliography, mathematical notation

Language

English (fluent), Japanese (native speaker)

CERTIFICATES

Human Subjects Research (HSR) Training for Social & Behavioral Investigators, Collaborative Institutional Training Initiative (CITI Program) 2026 - 2029
UC Irvine Certified Learning Assistant Program (CLAP) 2024
MySQL for Data Analytics and Business Intelligence, Udemy 2021